CITY MANAGEMENT

Request For Information

Unit and Tenant Information		
To: Tenant Fit Out	RFI#	HMG-KAFD-RFI-ELEC-003
	Date	28/10/2025
Tenant.fitout@kafd.sa	Tenant name	HMG
	Building	2.12
	and Unit #	2,3,4,5

RFI Description (Electrical Supply for X-ray Machine – Utilization of Spare Breaker in SMDB-01-03)

Concerning the electrical supply for the X-ray machine, a site survey of the LV room was carried out jointly by HMG and Samman design teams.
It is recommended to utilize the **spare 160A circuit breaker in SMDB-01-03** to supply power to the X-ray machine, making use of the existing **KAFD cable tray**.
According to the enhancement drawings for the low voltage distribution schematic diagram (**1110-2.12-GEN-SD-E-EP-615101/5**) and the electrical containment layout (**1110-2.12-L01-SD-E-EP-615207**), **SMDB-01-03** exclusively feeds **Dr. Sulaiman Al-Habib Medical Center**.
The current feeders are as follows:

- **Feeder (L01-18A)** with 160A C.B supplies **Retail 5 (Laboratory)**.
- **Feeder (0101-6R4 G31)** supplies **Retail 3 (Pharmacy)**.
- **Feeder (L01-10A)** supplies **Retail 2 (Dental X-ray)**.

Previously, the disconnect switch in **Retail 2** was not available. However, after its installation yesterday, the maintenance team confirmed in our presence that this feeder indeed serves **Retail 2**.
Please find attached the **calculation report for SMDB-01-03**, which confirms its adequacy to feed the X-ray machine through the available **160A spare breaker**.

Attachments:

Panel load calculation	Photographs of L.V Panel in electric room in first floor	Enhancement drawings	

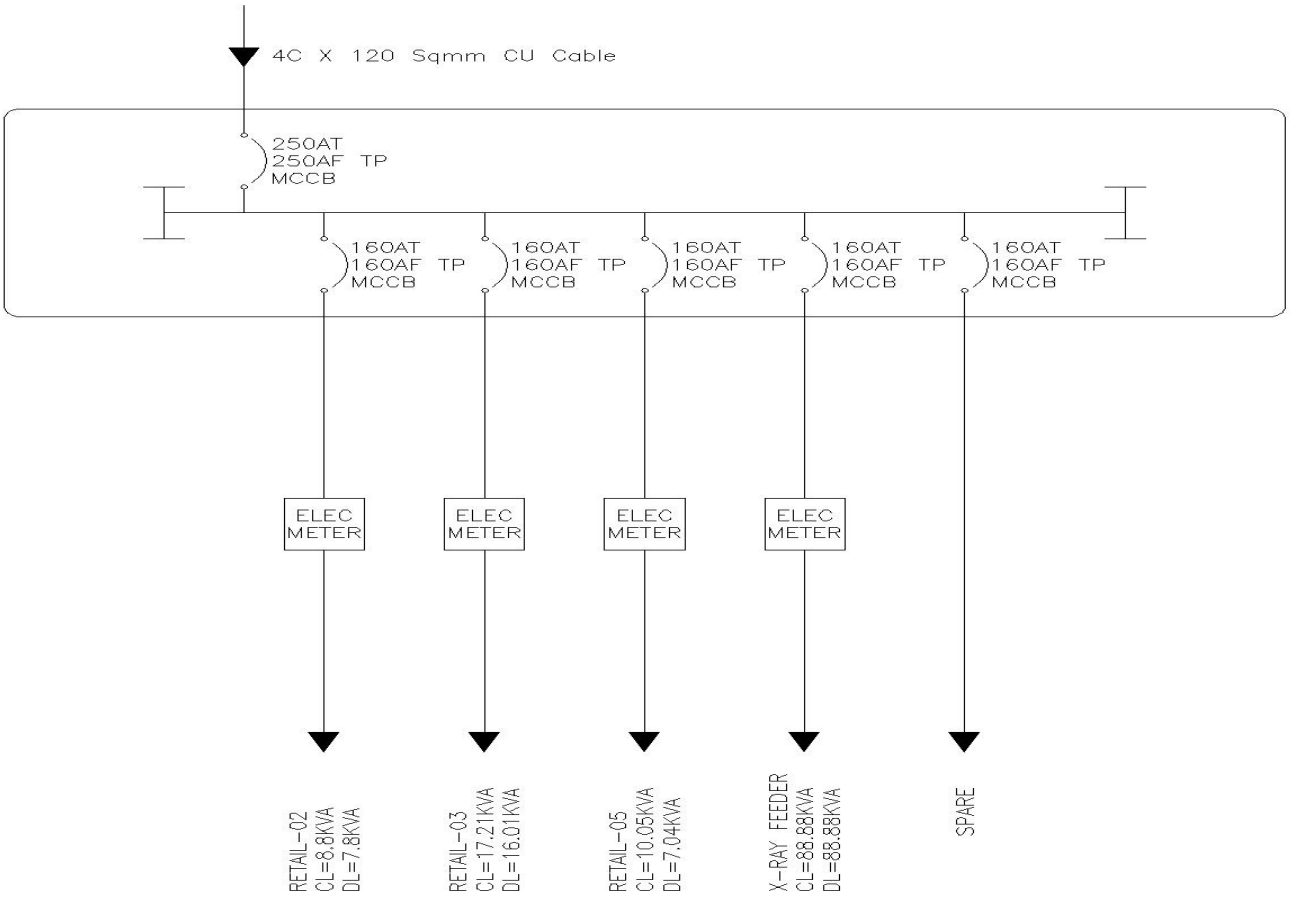
Submitted By:

Eng. Abdelrahman Kormod	HMG		

Response To RFI (KAFD DMC)

Response By:	Name, Title	Date	Click or tap to enter a date.
	Company		

Electrical Supply for X-ray Machine – Utilization of Spare Breaker in SMDB-01-03
PANEL LOAD CALCULATION



1. Actual Load of Retail-02

   Connected Load = 8.8 KVA

   Demand Load = 7.8 KVA

2. Actual Load of Retail-03

   Connected Load = 17.21 KVA

   Demand Load = 16.01 KVA

3. Actual Load of Retail-05

   Connected Load = 10.05 KVA

   Demand Load = 7.04 KVA

- Total Connected Load = 8.8+17.21+10.05 KVA
  = 36.06 KVA

- Total Connected Load = 8.8+17.21+10.05 KVA
  = 36.06 KVA
- The Required Load for X-Ray = 80 KW @ 0.9 P.F. = 80/0.9 = 88.88 KVA

- So, the total required load after connecting the X-Ray feeder from the same panel SMDB-01-03 comes out to be.

  Connected Load = 124.94 KVA

  Demand Load = 119.73 KVA

  The required protection for the new demand load 119.73 KVA

= Apparent Power /(√3 x System Voltage) = New Demand Load /(√3 x System Voltage) = 119.73/(√3 x380)

= 182 A

- If applying 20% safety factor = 218.29 A. So, its safe because rating of Incoming breaker = 250 A
- Verifying the protection of Individual X-Ray Feeder

  I = 88.89 / (√3 x 380) = 135.04 A

  Considering 20% derating of Breaker = 160 A (Approx) which is safe.
- The Incoming cable size as mentioned in SLD is 4C x 150 mm^2 but on site found that the size of cable is 4C x 120 mm^2.
- The current carrying capacity of 4C x 120 mm^2 cable is approx. 300 A in Air medium which is safe as after connecting the X-Ray feeder load is not going beyond 250 A.

PANEL BOARD SMDB 01-03

MAIN
Compact
INS 250
Schneider
Electric
I
ON
O
OFF
Ui 800V
Uimp 8kV
Ith 250A/60°C
50/60 Hz
AC22A AC23A DC22A
Ue (V) 690 690 250 (4P)
Ie (A) 250 250 250
IEC / EN 60947-3

MAIN C.B 250A

PANEL FROM INSIDE

SUB MAIN C.BS 160A FOR RETAIL 2,5

SPARE C.B 160 A & C.B 160A FOR RETAIL 3